Dataset

Ramdom Examples → Gemini / OpenAI → Synthesized Examples → Filtering

LOOP

Missing Necessary Tools | Potential Tools | Limited Functionality Tools

→ Unsolvablized

Unsolvable - MNT | Unsolvable - PT | Unsolvable - LFT

Unsolvable Task Example in PT Scenario

Unsolvable User Query	Unsolvable Tool List
I want to know: 1. What ports are available on the computer. 2. What program or process they are occupied by. 3. And generate a report.	**PortLister** Lists all ports on the computer. **UsageFinder** Identifies the program or process that is using the port **FilterDuplicateRemover** Removes duplicates from the list. **Finish** Used to end the task. **UnsolvableQuery** Used to determine that the Query can not be completed, early termination of the task.

Evaluation

1. Solvablility Detection
2. Solution Planning
3. Missing-Tool Analysis

→ LLM

Exact Match

Predicted Solvability ⇄ Ground Truth Solvability

Progress Rate

Predicted tool planning ⇄ 

Progress Rate

Predicted subgoal planning — P-ability 3 ⇄

Subgoal 1 —【describe 1】— Planned

Subgoal 2 —【describe 2】— Planned

Subgoal 3 —【ability 3】— Planned

Match Score P-ability 3 ⇄ ability 3

- If exact match found score = 1.0.
- Otherwise, compute cosine similarity.

$$cos_{sim}(A,B) = \frac{A \cdot B}{||A||||B||}$$

Hallucination Diagnosis

